



THE MISSION: SAMPLING

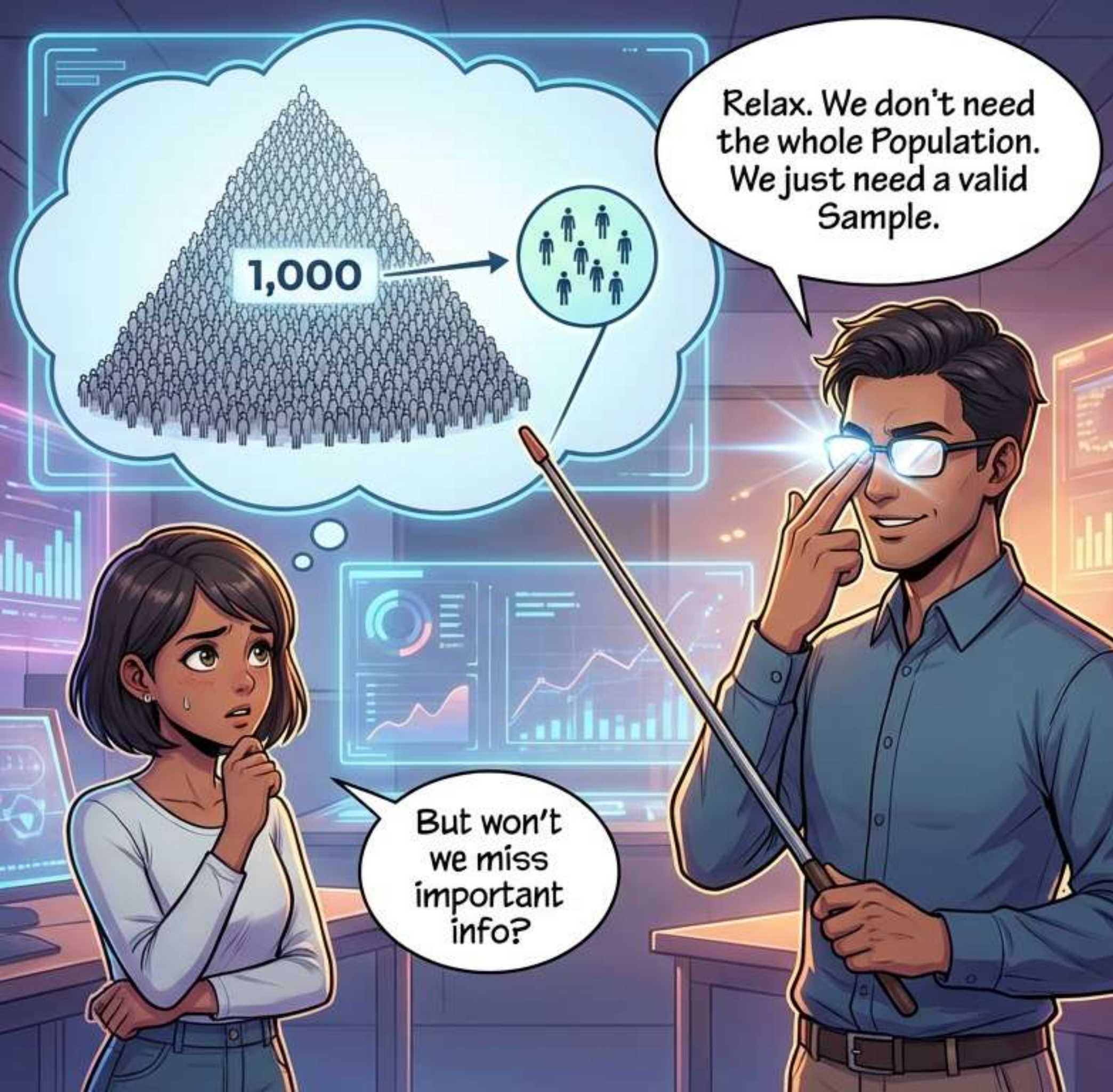
Goal: Conduct research on a specific group without asking every single person.

The Problem: Collecting data from an entire population is rarely possible due to time, cost, and accessibility.

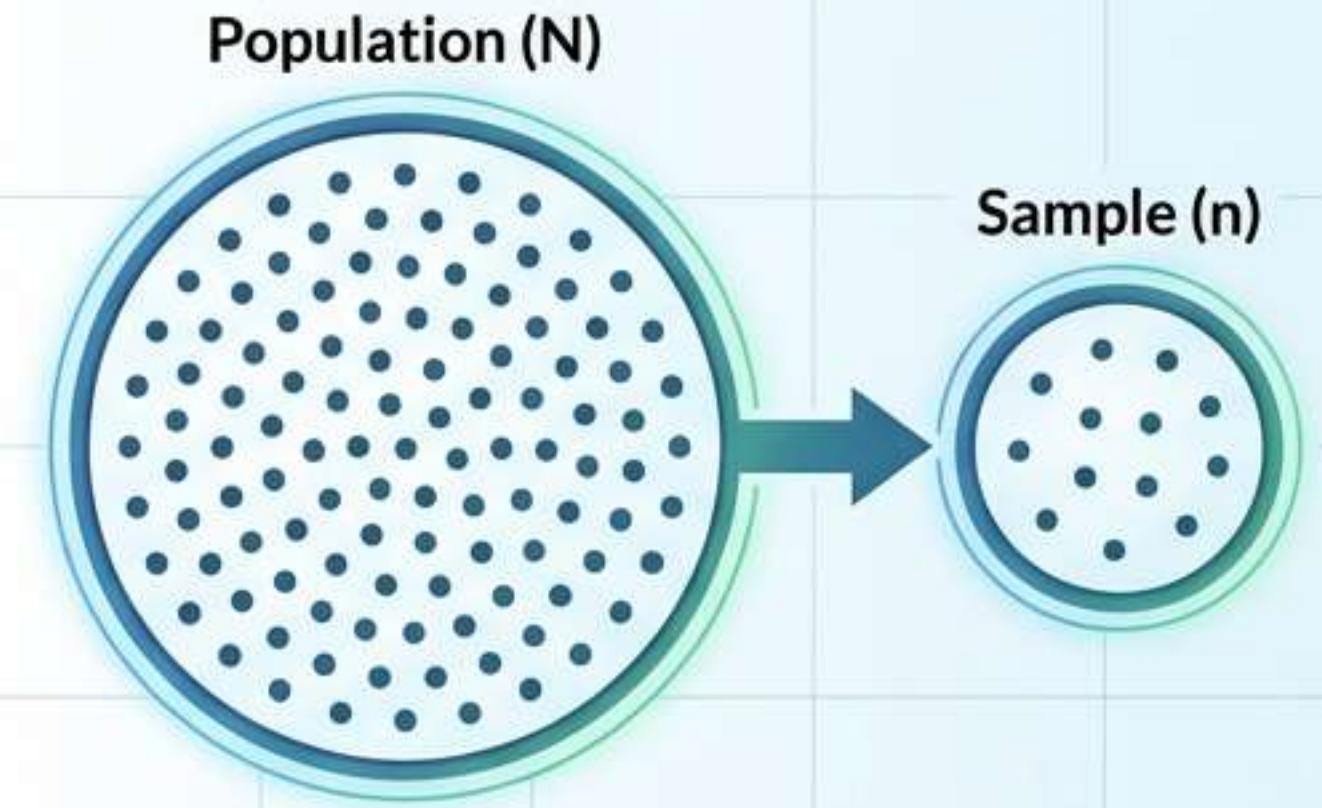


The Solution: Select a **Sample**—a smaller group of individuals who will actually participate in the research to represent the whole.





POPULATION VS. SAMPLE



- **Population:** The entire group you want to draw conclusions about.
 - *Example:* All 1,000 employees of the company.
- **Sample:** The specific group of individuals you collect data from.
 - *Example:* 100 employees selected for the survey.
- **Key Takeaway:** The population can be broad (adults in a country) or narrow (students in a single school).



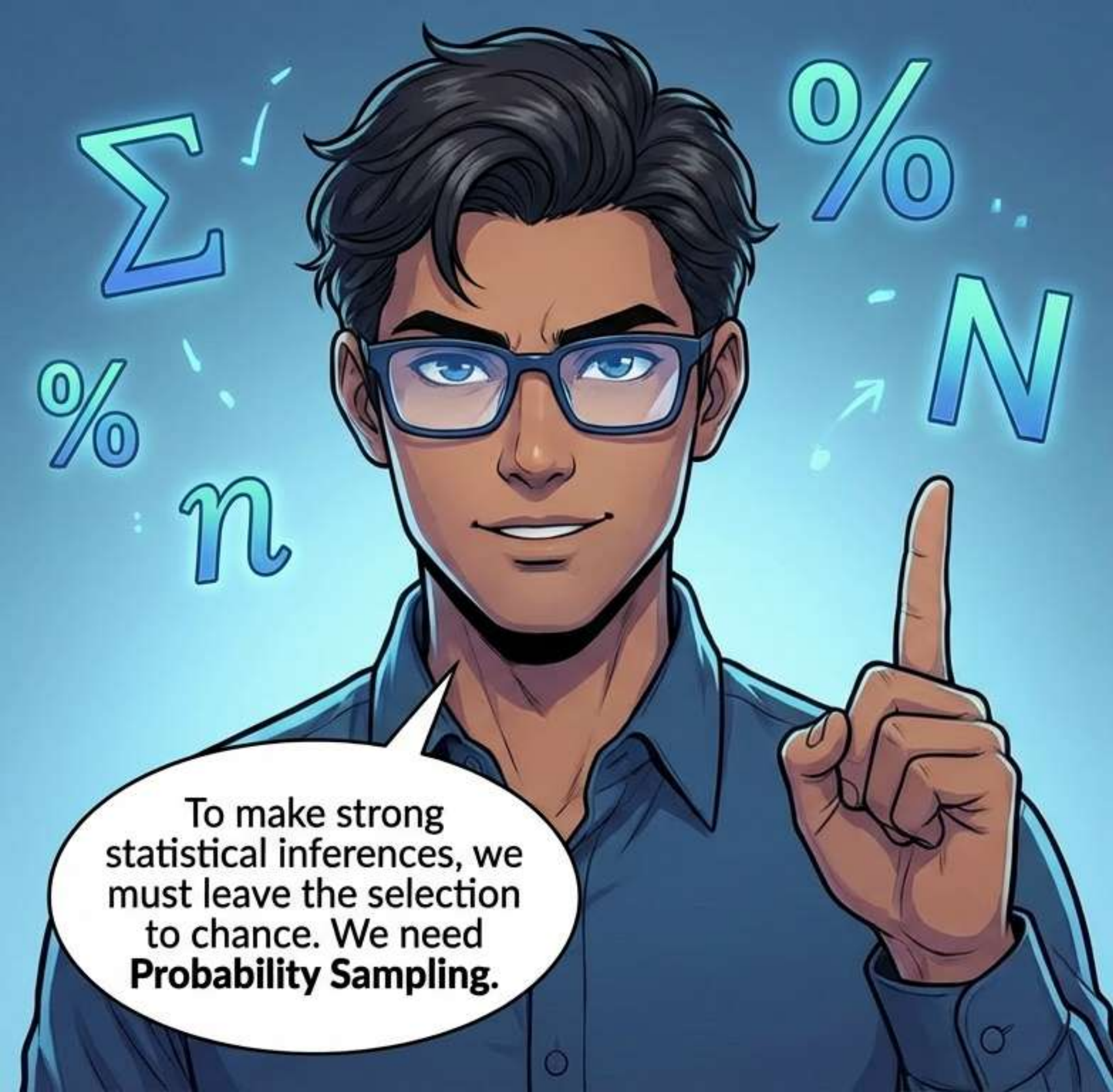
THE SAMPLING FRAME

Definition: The actual list of individuals that the sample will be drawn from.

The Ideal: The frame should match the target population perfectly.

Real-World Example:

- **Target Population:** All employees.
- **Sampling Frame:** The HR database containing 1,000 entries.
 - **Note:** If the list is outdated, you risk **Coverage Error**.



To make strong statistical inferences, we must leave the selection to chance. We need **Probability Sampling**.

ACT II: PROBABILITY SAMPLING

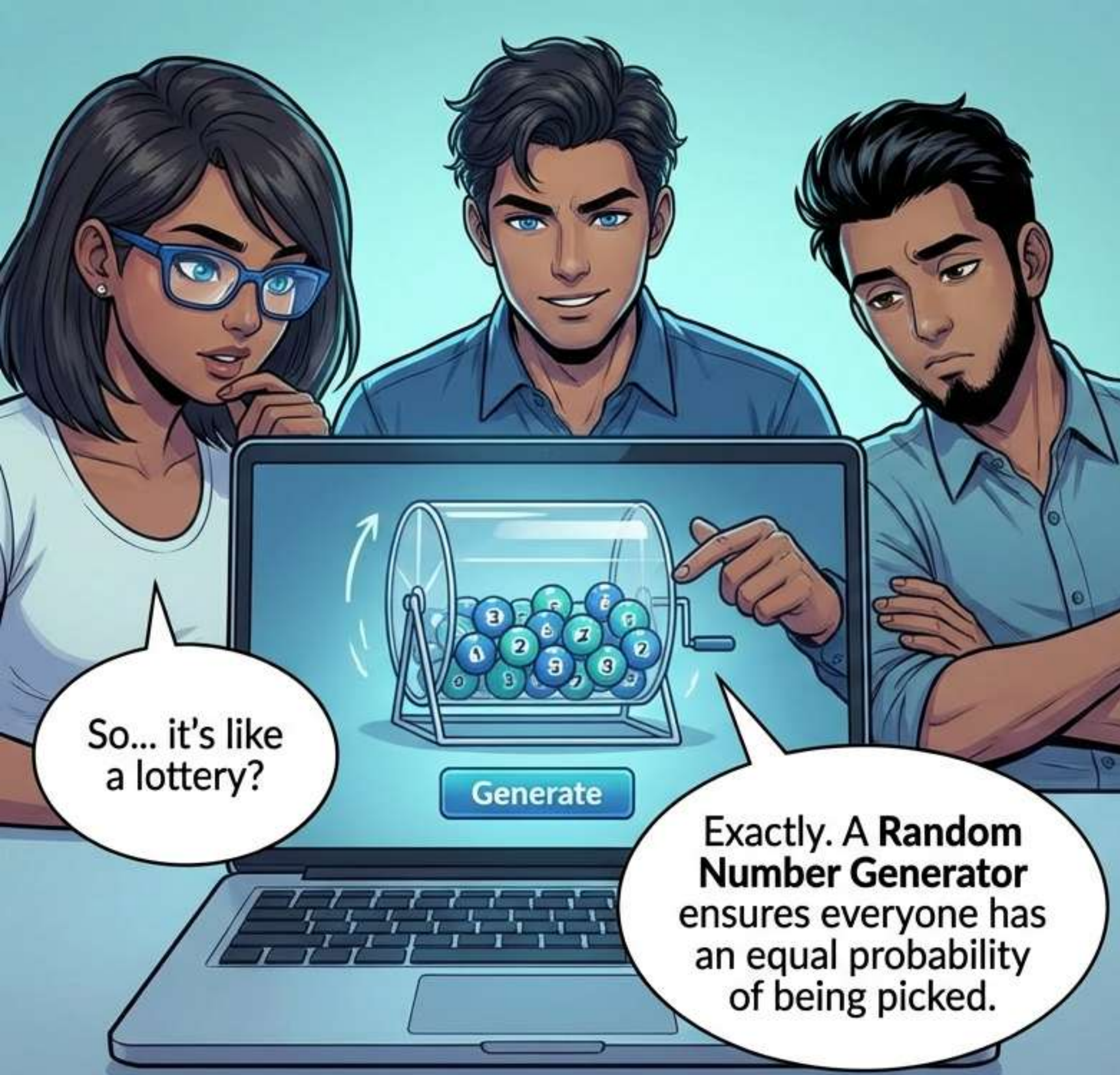
Core Concept: Every member of the population has a known, non-zero chance of being selected.

Primary Use: Quantitative research.

Benefit: Produces results that are representative of the whole group.

The 4 Main Techniques:

1. Simple Random Sampling
2. Systematic Sampling
3. Stratified Sampling
4. Cluster Sampling



1. SIMPLE RANDOM SAMPLING

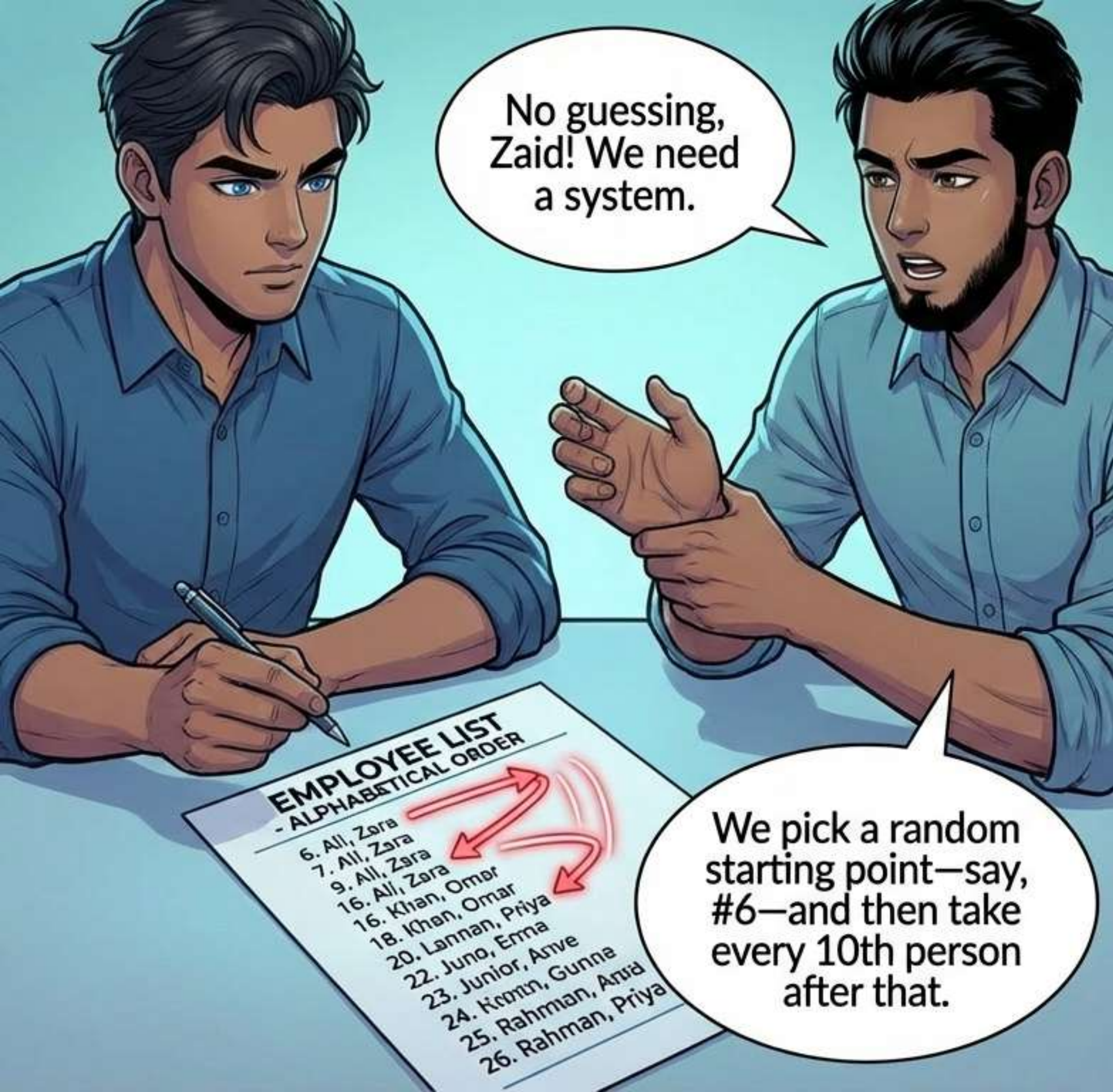


Method: Every member of the population has an equal chance of selection.

The Process:

1. Assign a number to every employee (1–1000).
2. Use a random number generator to select 100 numbers.

Requirement: The sampling frame must include the *entire* population.



2. SYSTEMATIC SAMPLING

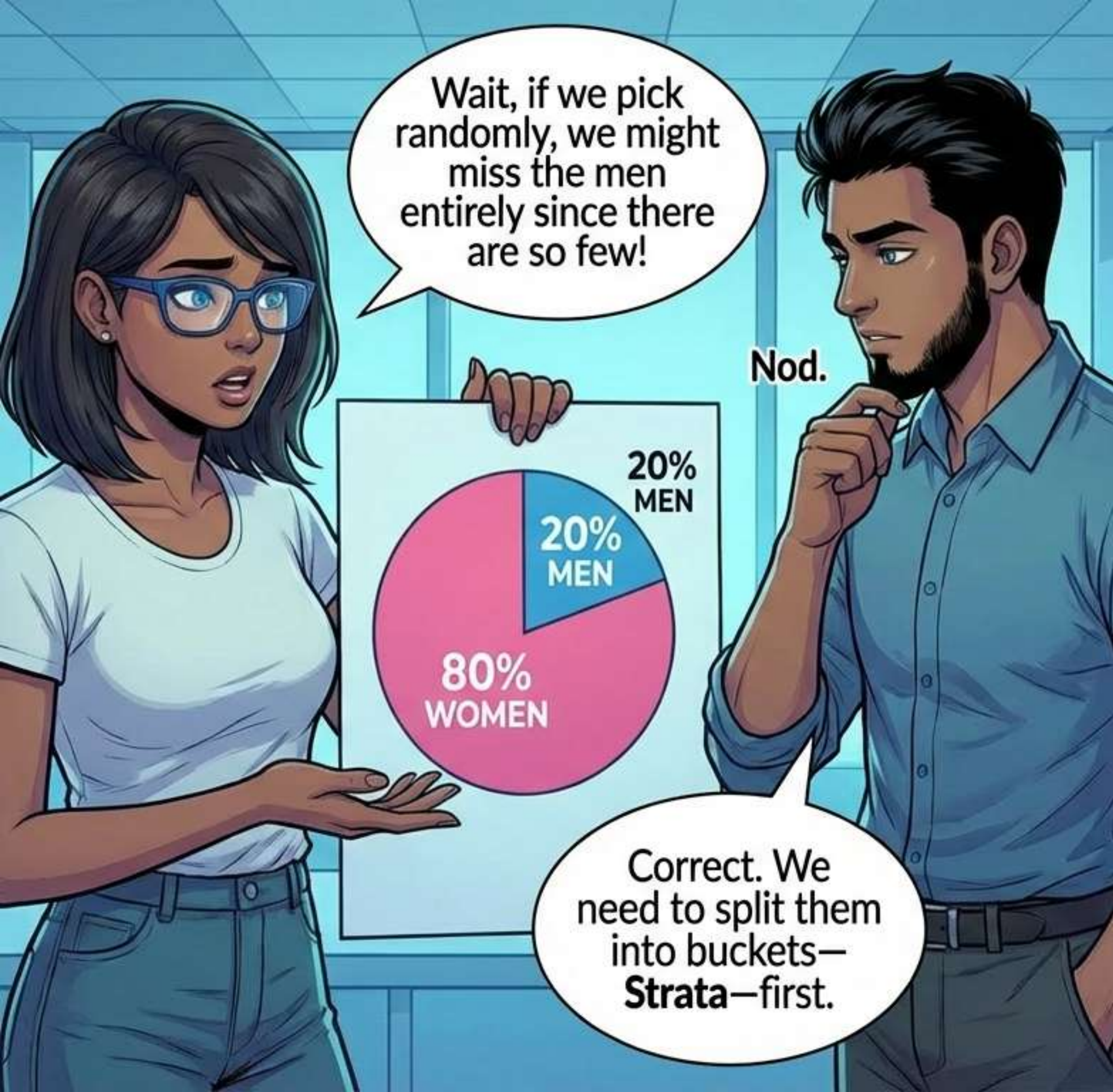


Method: Individuals are chosen at regular intervals.

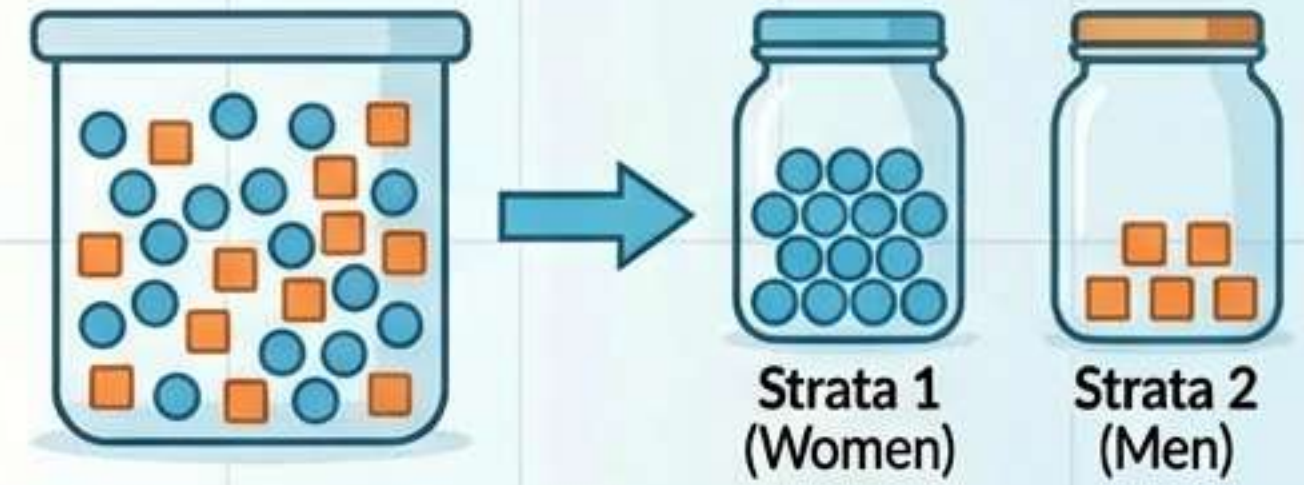
The Process:

1. List the population (e.g., alphabetical order).
2. Pick a starting point (e.g., #6).
3. Select every 10th person (6, 16, 26, 36...).

Risk: Watch for hidden patterns! If the list is ordered by seniority (Manager, Senior, Junior, Manager...), skipping every 3rd person might result in only picking Managers.



3. STRATIFIED SAMPLING



Method: Divide population into subpopulations (strata) that differ in important ways (e.g., gender, age, income).

Why: Ensures every subgroup is properly represented.

The Example:

- **Strata 1 (800 Women)::** Randomly select 80.
- **Strata 2 (200 Men)::** Randomly select 20.
- **Total Sample:** 100 people (Reflecting the exact 80/20 split).



4. CLUSTER SAMPLING



Method: Divide population into subgroups (clusters). Randomly select entire subgroups.

The Process:

1. Identify Clusters (The 10 Offices).
2. Randomly select 3 Offices.
3. Survey every employee in those 3 offices.

Trade-off: Good for large/dispersed populations, but higher risk of error if the selected offices differ substantially from the others.



ACT III: NON-PROBABILITY SAMPLING

Core Concept: Individuals are selected based on non-random criteria (convenience or judgment).

Primary Use: Exploratory and qualitative research.

Trade-off: Easier and cheaper to access, but higher risk of **Sampling Bias**.

Validity: Inferences about the population are weaker than with probability samples.



1. CONVENIENCE SAMPLING

Method: Includes individuals who happen to be most accessible to the researcher.

Pros: Easy and inexpensive way to gather initial data.

Cons: No way to tell if the sample is representative. High risk of selection bias.

Example: Surveying only students in your own classes (biases results toward your specific major/level).



2. VOLUNTARY RESPONSE SAMPLING



Method: Participants volunteer themselves (e.g., responding to a public online survey).

Bias Risk: Self-Selection Bias.

The Reality: People with strong opinions (very happy or very angry) are most likely to respond, skewing the 'average' opinion. The moderate majority stays silent.



3. PURPOSIVE (JUDGMENT) SAMPLING

Method: The researcher uses their expertise to select a sample most useful to the research purpose.

Use Case: Qualitative research or very small/specific populations.

Example: Deliberately selecting students with different physical and learning support needs to gather detailed data on accessibility services.

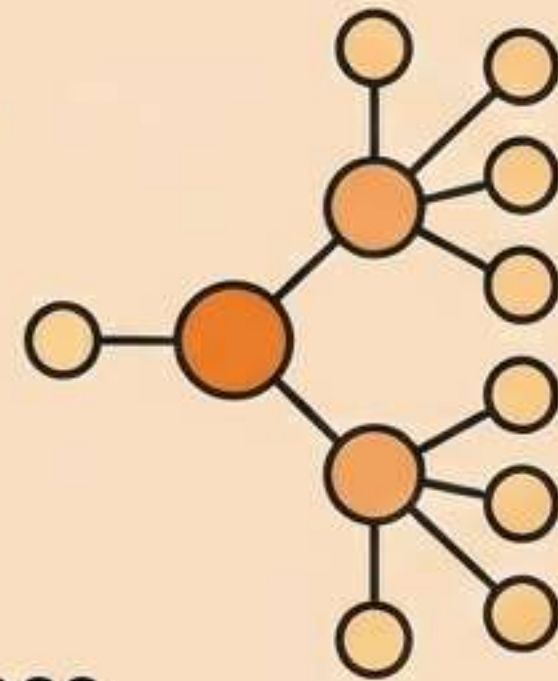
Note: Requires clear criteria for inclusion.



What about hard-to-reach groups? Like homeless students? There's no list for that.

We find one person, build trust, and ask them to introduce us to others.

4. SNOWBALL SAMPLING



Method: Recruitment via other participants.

Use Case: Hard-to-access populations (e.g., homelessness, underground communities).

Process:

1. Meet one participant.
2. They put you in contact with others they know.
3. The sample 'snowballs' in size.

Cons: Representativeness is unknown; relies heavily on social networks.



Meat Eaters: 200/200 (FULL)

Vegetarians: 150/200

Vegans: 10/200



Sorry! We have enough meat eaters. I'm looking for Vegans now!

Bilal: He's filling the Quota manually.

5. QUOTA SAMPLING

Method: Non-random selection of a predetermined number (quota) of units.

Process:

1. Divide population into strata (e.g., Diet: Meat, Veg, Vegan).
2. Set a quota (e.g., 200 of each).
3. Recruit until you fill the quota.

Goal: Ensures small groups are represented, but selection within the groups is not random.

 **BILAL**
PROBABILITY

VS

 **ZAID**
NON-PROBABILITY



THE SHOWDOWN: WHICH IS BETTER?

PROBABILITY (Bilal)	NON-PROBABILITY (Zaid)
Random Selection	Non-Random Selection
Strong statistical inferences	Weaker inferences
Representative of the whole	High risk of Bias
Harder / More Expensive	Easier / Cheaper
Best for: Generalization	Best for: Exploratory Insight



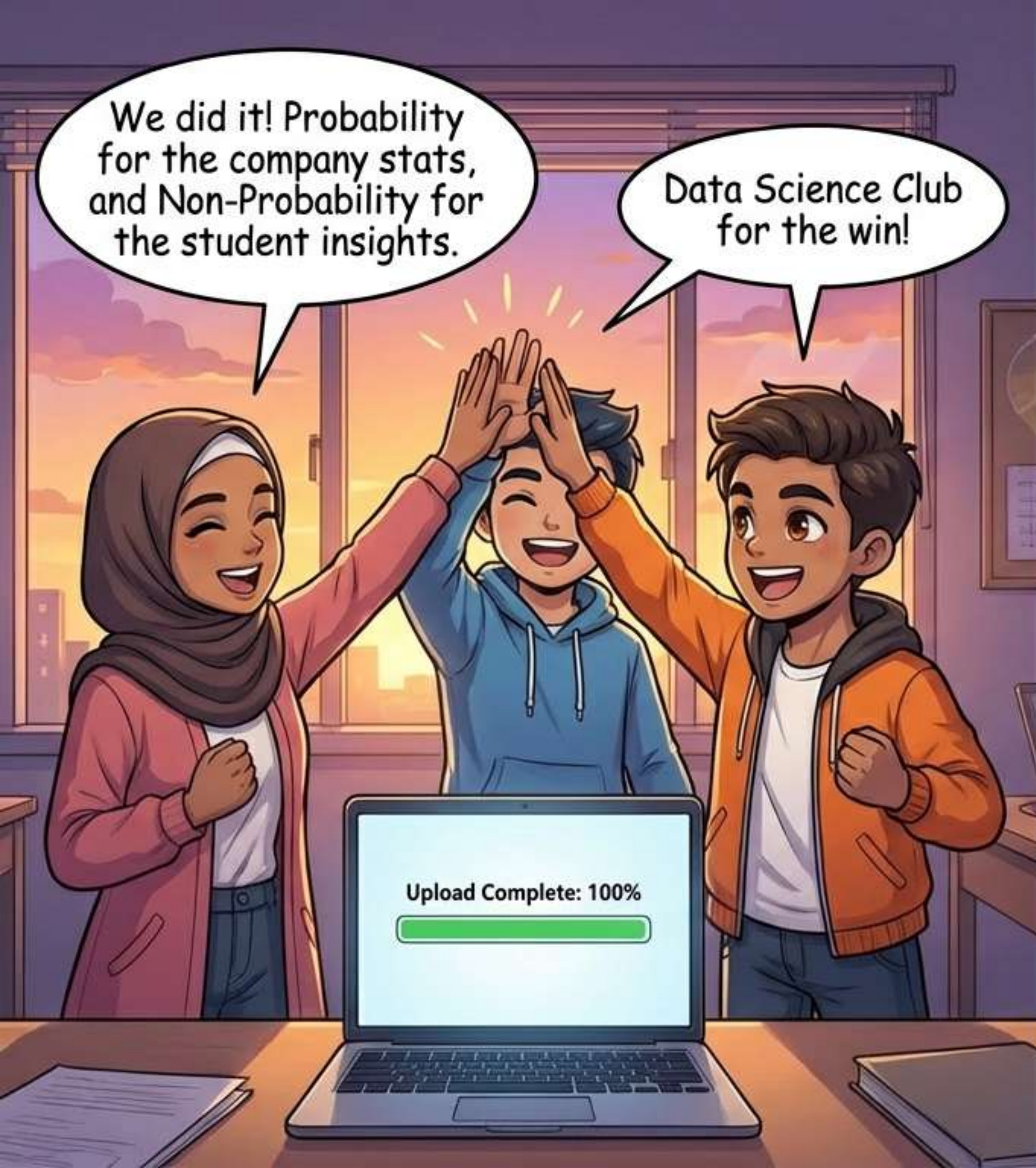
BEWARE: SAMPLING BIAS

Definition: When some members of a population are systematically more likely to be selected than others.

Consequence: You cannot generalize your findings to the population.

Prevention:

1. Use Probability sampling when possible.
2. Clearly explain your sampling method in your methodology section.
3. Discuss how you minimized bias.



FINAL LESSON

1. **Determine your goal:** Do you need **Generalization** (Probability) or **Insight** (Non-Probability)?
2. **Define your Population and Sampling Frame** early.
3. **Choose the method** that fits your resources, timeline, and research questions.

**Source: McCombes, S. (2023). Sampling Methods. Scribbr.*